

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

MERN STACK
Practical – II

Roll No.: T032	Name: KHUSHI
Class: TYIT	Batch: 2
Date of Assignment: 30-06-26	Date/Time of Submission: 30-06-26

a. Create functions and call them with parameters.

CODE:-

For calculateSalary :-

```
function calculateSalary(name, basicSalary) {
  let bonus = 5000;
  let totalSalary = basicSalary + bonus ;
  console.log ("Employee Name : ", name);
  console.log ("Basic Salary : ", basicSalary );
  console.log ("Bonus: ", bonus);
  console.log ("Total Salary : ", totalSalary);
}
//function call
calculateSalary("Rohan", 30000);
```

For generating bill with 18% GST :-

```
function genBill(Product,Quantity,Price) {
  let total = Quantity*Price;
  let bill = total+(total*0.18) ;

  console.log ("Product: ", Product);
  console.log ("Quantity : ", Quantity );
  console.log ("Total bill : ", bill);
  console.log ("Price: ", Price);
}
//function call
genBill("Rohan",1,30000);
console.log("Khushii T032");
```

OUTPUT:-

```
PS C:\Users\info\OneDrive\Desktop\mern prac> node prac2a.js
Employee Name :  Rohan
Basic Salary :  30000
Bonus:  5000
Total Salary :  35000
Khushii T032

Node.js v24.17.0
PS C:\Users\info\OneDrive\Desktop\mern prac> node prac2a.js
Product:  Rohan
Quantity :  1
Total bill :  35400
Price:  30000
Khushii T032
```

b. Work with arrays and objects in JavaScript.

CODE:-

//arrays

```
let fruits = ["Mango" , "Banana" , "Apple" , "Cherry"];
```

//object

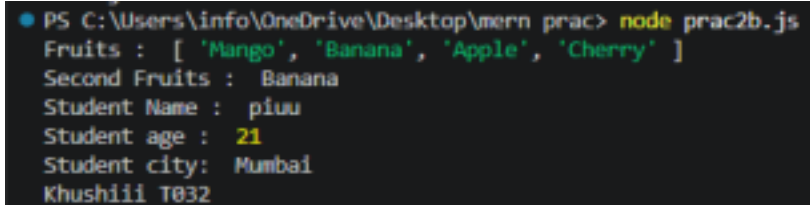
```
let student = {
  name: "piuu",
```

```

age : 21 ,
city: "Mumbai"
};
console.log("Fruits : " , fruits );
console.log("Second Fruits : " , fruits[1]);
console.log("Student Name : " , student.name);
console.log("Student age : " , student.age);
console.log("Student city: " , student.city);
console.log("Khushiii T032 ");

```

OUTPUT:-



```

PS C:\Users\info\OneDrive\Desktop\mern prac> node prac2b.js
Fruits : [ 'Mango', 'Banana', 'Apple', 'Cherry' ]
Second Fruits : Banana
Student Name : piuu
Student age : 21
Student city: Mumbai
Khushiii T032

```

c. Demonstrate use of arrow functions.

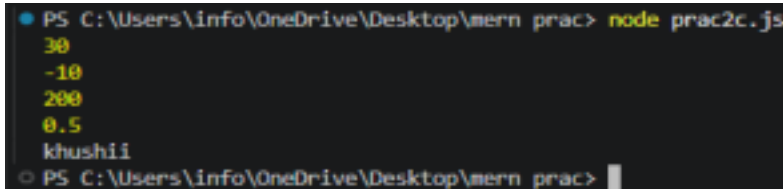
CODE:-

```

let add = (a , b) => {
  return a + b ;
};
let sub = (a , b) => {
  return a - b ;
};
let mul = (a , b) => {
  return a * b ;
};
let div = (a , b) => {
  return a / b ;
};
console.log(add(10,20));
console.log(sub(10,20));
console.log(mul(10,20));
console.log(div(10,20));
console.log("khushii");

```

OUTPUT:-



```

PS C:\Users\info\OneDrive\Desktop\mern prac> node prac2c.js
30
-10
200
0.5
khushii
PS C:\Users\info\OneDrive\Desktop\mern prac>

```

d. Write a program using array methods (map, filter, reduce).

CODE:-

```

let numbers = [10 , 20 , 30 , 40 , 50 ];
// map()
let doubled = numbers.map(num => num * 2 );
//filter()
let greater25 = numbers.filter(num => num > 25 );
// reduce()
let total = numbers.reduce((sum,num) => sum + num , 0 );
console.log ("Oiginal Array : " , numbers );
console.log ("Map : " ,doubled);
console.log("filter : " , greater25);
console.log("Reduce : " , total);
console.log("Khushii");

```

OUTPUT:-

```

PS C:\Users\info\OneDrive\Desktop\mern prac> node prac2d.js
Original Array : [ 10, 20, 30, 40, 50 ]
Map : [ 20, 40, 60, 80, 100 ]
filter : [ 30, 40, 50 ]
Reduce : 150
Khushii
PS C:\Users\info\OneDrive\Desktop\mern prac>

```

e. Create a program that manipulates objects and displays output.

CODE:-

//1.Creating an Object

```

let student = {
  rollNo : 32,
  name : "kuhuu",
  marks : 75
};

```

//2. Accessing Properties

```

console.log("\nAccessing Properties : ");
console.log("Name : ", student.name);
console.log("Marks : ", student.marks);

```

//3. Updating a Property

```

student.marks = 90;
console.log ("\nAfter updating Marks : ");
console.log(student);

```

//4.Adding a new Property

```

student.city = "Mumbai" ;
console.log ("\nAfter adding city : ");
console.log(student);

```

//5. deleting a property

```

delete student.rollNo;
console.log("\nAfter Deleting roll number : ");
console.log(student);
console.log("Khushii T032");

```

OUTPUT:-

```

PS C:\Users\info\OneDrive\Desktop\mern prac> node prac2e.js

Accessing Properties :
Name : kuhu
Marks : 75

After updating Marks :
{ rollNo: 32, name: 'kuhu', marks: 90 }

After adding city :
{ rollNo: 32, name: 'kuhu', marks: 90, city: 'Mumbai' }

After Deleting roll number :
{ name: 'kuhu', marks: 90, city: 'Mumbai' }
Khushii T032

```

f. Implement a small JavaScript program combining functions, arrays, and objects.

CODE:-

// Array of objects

```

let students = [
  {
    rollno : 11,
    name : "Bob",
    marks : 45
  },
  {
    rollno : 12,
    name : "Macro",

```

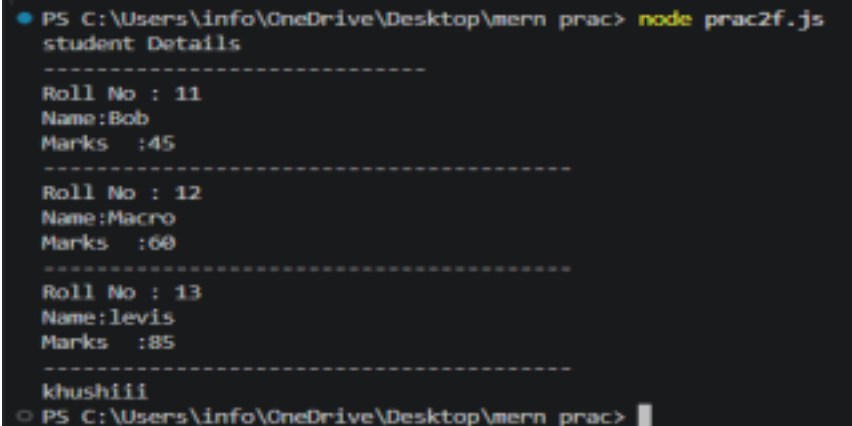
```

    marks : 60
  },
  {
    rollno : 13,
    name : "levis",
    marks : 85
  }
];

//Function to display Student Details
function display(s)
{
  console.log("student Details");
  console.log("-----");
  for (let stud of s )
  {
    console.log(`Roll No : ${stud.rollno}`);
    console.log(`Name:${stud.name}`);
    console.log(` Marks :${stud.marks}`);
    console.log(`-----`);
  }
}

//function call
display(students);

```

OUTPUT:-


```

PS C:\Users\info\OneDrive\Desktop\mern prac> node prac2f.js
student Details
-----
Roll No : 11
Name:Bob
Marks :45
-----
Roll No : 12
Name:Macro
Marks :60
-----
Roll No : 13
Name:levis
Marks :85
-----
khushiii
PS C:\Users\info\OneDrive\Desktop\mern prac>

```